

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/380911535>

The sun's position at birth is unrelated to subjective well-being: Debunking astrological claims

Article in *Kyklos* · May 2024

DOI: 10.1111/kykl.12395

CITATIONS

0

READS

2,436

1 author:



[Mohsen Joshanloo](#)

Keimyung University

189 PUBLICATIONS 5,012 CITATIONS

[SEE PROFILE](#)

The sun's position at birth is unrelated to subjective well-being: Debunking astrological claims

Mohsen Joshanloo 

Department of Psychology, Keimyung University, Daegu, South Korea

Correspondence

Mohsen Joshanloo, Department of Psychology, Keimyung University, 1095 Dalgubeol Boulevard, Dalseo-Gu, Daegu 42601, South Korea.
Email: mjoshanloo@gmail.com

Abstract

Beliefs linking zodiac signs to personality traits, life outcomes, and well-being remain widespread across various cultures. This study examined the relationship between Western zodiac signs and subjective well-being in a nationally representative American sample from the General Social Survey ($N = 12,791$). Well-being was measured across eight components: general unhappiness, depressive symptoms, psychological distress, work dissatisfaction, financial dissatisfaction, perceived dullness of one's life, self-rated health, and unhappiness with marriage. Parametric and nonparametric analyses consistently revealed no robust associations between zodiac signs and any of the well-being variables, regardless of whether demographic factors were controlled for. The effect sizes were negligible, accounting for 0.3% or less of the variance in well-being, demonstrating that zodiac signs lack predictive power for well-being outcomes. An additional analysis revealed that astrological signs were no more predictive of than random numbers. Thus, a randomly generated number between 1 and 12 is statistically as predictive of one's well-being as one's zodiac sign. These findings challenge popular astrological claims about the influence of zodiac signs on well-being and quality of life.

1 | INTRODUCTION

Astrology is based on the premise that the positions and movements of celestial bodies at the time of birth shape personality, behavior, and life outcomes (Das et al., 2022). In Western astrology, the zodiac is divided into 12 signs

that correspond to different constellations along the sun's path from Earth's perspective. The signs are Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius, and Pisces. According to astrological belief, each sign is associated with certain personality traits and life outcomes. Astrology posits that zodiac signs influence one's personality traits and life path (LaBrot et al., 2022).

The belief that celestial bodies influence human affairs has been prevalent across various cultures and historical periods and continues to persist in the contemporary world. For example, about 30% of Americans believe in astrology (Pew Research Center, 2018). The scientific community has raised concerns about the validity of astrological claims (e.g., McGrew & McFall, 1990). Early empirical studies investigating the relationship between astrological signs and real-life outcomes have generally failed to find support for astrological claims (Clarke et al., 1996; Veno & Pamment, 1979). However, some of the earlier studies have faced criticism for methodological flaws such as small sample sizes (LaBrot et al., 2022). More recent research has attempted to overcome the limitations of earlier studies by employing larger sample sizes, using more advanced statistical methods, and expanding the conceptual scope of the investigations. For instance, Helgertz and Scott (2020) analyzed a large Swedish dataset and found no evidence to support the idea that astrologically compatible couples are more likely to marry or less likely to divorce. Similarly, Lu et al. (2020) used large Chinese samples to investigate the relationship between astrological signs and personality traits and job performance but found no significant correlations.

Pertinent to the present study, astrology systems claim astrological signs are linked to well-being. For example, the article titled “Astrological Signs Skilled In Maintaining Happiness¹” argues that “certain zodiac signs ... are really good at keeping away negative feelings. These signs have special qualities that help them stay positive and create a happy atmosphere.” Similarly, an article titled “5 Zodiac Signs Who Find It Easy To Be Happy²” suggests that certain zodiac signs are more prone to feeling happy. At present, there is a shortage of empirical evidence to either substantiate or challenge the claims about the relationship between astrological signs and subjective well-being. Despite the widespread prevalence of astrological beliefs and the numerous claims made about the influence of zodiac signs on happiness and well-being, no previous empirical research has comprehensively examined the relationship between zodiac signs and various facets of subjective well-being. Consequently, there is a need for more research to explore and illuminate this understudied link.

On what basis could zodiac signs be associated with well-being? Popular belief suggests that certain zodiac signs are associated with superior or inferior well-being outcomes through unexplained supernatural forces. Fidrmuc and De Dios Tena (2015), on the other hand, propose two psychosocial mechanisms for the potential influence of zodiac signs on life outcomes. The first mechanism posits that people who believe that zodiac signs influence health and well-being may consciously or unconsciously act in ways that confirm those beliefs. Consequently, individuals may end up feeling happier or unhappier. The second mechanism suggests that knowledge of one's zodiac sign within social circles may lead to differential treatment and judgment based on astrological perceptions. That is, people's well-being could be affected by how others perceive and interact with them based on their zodiac sign. The first and second mechanisms may be titled “self-fulfilling influence” and “astrological stereotyping,” respectively. These potential pathways underscore the possibility that knowledge of zodiac signs may trigger real-life mechanisms, which could ultimately have an impact on different facets of well-being. Empirical evidence spanning various disciplines has indeed demonstrated the significant impact that perceptions of one-self and others can have on life outcomes (Romney et al., 2023). Thus, even if we refute the idea of supernatural influence from astrology on human life, it is still conceivable that astrological signs might affect life outcomes through various psychosocial mechanisms. Given the importance of evidence-based understanding, there is a need for empirical studies that can either confirm or refute the supposed impact of astrological signs on individuals' lives.

¹<https://timesofindia.indiatimes.com/astrology/zodiacs-astrology/astrological-signs-skilled-in-maintaining-happiness/articleshow/107180526.cms>

²<https://timesofindia.indiatimes.com/life-style/relationships/love-sex/5-zodiac-signs-who-find-it-easy-to-be-happy-and-those-who-dont/photostory/81312918.cms?from=mdr>

1.1 | The present study

Recognizing the need to fill the existing empirical gap, this study sought to examine the potential associations between Western zodiac signs and various dimensions of subjective well-being. To accomplish this, a wide range of well-being components were included, such as satisfaction with physical health, general happiness, marital satisfaction, financial satisfaction, and job satisfaction. By including such a diverse set of well-being measures, the study aimed to provide a comprehensive understanding of the possible relationships between astrological signs and subjective well-being. By utilizing a large, nationally representative sample of American adults and employing rigorous statistical analyses, this study aimed to provide a fair and robust assessment of the relationship between astrological signs and subjective well-being. The primary objective of this study was to provide an evidence-based examination of these purported relationships, contributing to a more informed public understanding of astrological signs and their impact on individual well-being.

2 | METHODS

2.1 | Participants

The data are from the General Social Survey (GSS), a nationally representative, cross-sectional survey that has been conducted in the United States since 1972. The survey collects data on a wide range of topics, including attitudes, opinions, and behaviors, in order to monitor trends and changes in American society over time. The GSS uses a full-probability sampling method. The data used in this study were drawn from the four most recent waves of the GSS, specifically the 2022, 2021, 2018, and 2016 waves. Table 1 provides an overview of the sample sizes and descriptive statistics for each wave. The total sample across all four waves consists of 12,791 participants, with an average age of approximately 50 (55% females). It is important to note that some responses are missing from the data and certain questions were not asked of all participants. As a result, the sample size for each individual analysis varies depending on the availability of data for the specific variables included in that analysis.

2.2 | Measures

This study has eight outcome variables. Table 2 presents all the variable names, items used to measure them, and their response formats. Table 3 presents the descriptive statistics for the variables.

- Zodiac sign. The GSS team derived the respondents' Western zodiac signs from their self-reported dates of birth.
- Depressive symptoms. Three items were used to measure depressive symptoms experienced in the past 4 weeks. A principal axis factor analysis was conducted to confirm the underlying factor structure of the scale. The results

TABLE 1 Descriptive statistics of the sample across waves.

Wave	N	Age	Male%	Female%
2022	3544	49.2	46.2%	53.8%
2021	4032	52.2	44.1%	55.9%
2018	2348	49.0	44.8%	55.2%
2016	2867	49.2	44.5%	55.5%
Total	12791	50.0	44.9%	55.1%

TABLE 2 Scales and items.

Variable	Wording	First and last anchors
General unhappiness	Taken all together, how would you say things are these days—would you say that you are very happy, pretty happy, or not too happy?	1 = <i>very happy</i> 3 = <i>not too happy</i>
Depressive symptoms	During the past 4 weeks how often ... • Have you felt unhappy and depressed? • Have you lost confidence in yourself? • Have you felt you could not overcome your problems?	1 = <i>never</i> 5 = <i>very often</i>
Distress	Over the last 2 weeks, how often have you been bothered by the following problems: • Feeling nervous, anxious, or on edge • Not being able to stop or control worrying • Feeling down, depressed, or hopeless • Little interest or pleasure in doing things	1 = <i>not at all</i> 4 = <i>nearly every day</i>
Work dissatisfaction	On the whole, how satisfied are you with the work you do—would you say you are very satisfied, moderately satisfied, a little dissatisfied, or very dissatisfied?	1 = <i>very satisfied</i> 4 = <i>very dissatisfied</i>
Dissatisfaction with financial situation	We are interested in how people are getting along financially these days. So far as you and your family are concerned, would you say that you are pretty well satisfied with your present financial situation, more or less satisfied, or not satisfied at all?	1 = <i>pretty well satisfied</i> 3 = <i>not satisfied at all</i>
Life dullness	In general, do you find life exciting, pretty routine, or dull?	1 = <i>exciting</i> 3 = <i>dull</i>
Poor health	Would you say your own health, in general, is excellent, good, fair, or poor?	1 = <i>excellent</i> 4 = <i>poor</i>
Unhappiness of marriage	(If currently married) taking things all together, how would you describe your marriage? Would you say that your marriage is very happy, pretty happy, or not too happy?	1 = <i>very happy</i> 3 = <i>not too happy</i>

TABLE 3 Descriptive statistics for variables (whole sample).

	N	Minimum	Maximum	Mean	Standard deviation
General unhappiness	12,737	1	3	1.956	.660
Depressive symptoms	1143	1	5	1.997	.994
Distress	1955	1	4	1.564	.700
Work dissatisfaction	8715	1	4	1.718	.805
Dissatisfaction with financial situation	12,736	1	3	1.976	.745
Life dullness	8429	1	3	1.614	.579
Poor health	11,018	1	4	2.106	.781
Unhappiness of marriage	5637	1	3	1.427	.569
Age	12,233	18	89	50.036	17.757
Education	12,693	0	20	14.162	2.929

suggested a unidimensional factor structure with the first factor explaining 65.728% of the variance. The initial eigenvalues for the first and second factors were 2.312 and 0.383, respectively. The factor loadings for the three items ranged from 0.766 to 0.847. The Cronbach's alpha coefficient for the scale was 0.851. This measure was only used in 2022.

- Psychological distress. Four items asked how often participants had symptoms of psychological distress during the past 2 weeks. A principal axis factor analysis was conducted to confirm the underlying factor structure of the scale. The results suggested a unidimensional factor structure with the first factor explaining 62.648% of the variance. The initial eigenvalues for the first and second factors were 2.875 and 0.561, respectively. The factor loadings for the three items ranged from .737 to .845. The Cronbach's alpha coefficient for the scale was .867. This measure was only used in 2022.
- Other variables. All other variables were measured with single items (see Table 3). Age, gender, and education (i.e., highest year of school completed) were used as control variables.

2.3 | Statistical analysis

JASP, SPSS, and R were used for data analysis. This study employed analysis of covariance (ANCOVA) as the primary method for hypothesis testing. ANCOVA is a statistical technique that allows researchers to examine differences between group means on a dependent variable while statistically controlling for the effects of covariates (Tabachnick & Fidell, 2019). ANCOVA models were tested separately for each outcome variable, with zodiac sign (a categorical variable with 12 categories) serving as the main independent variable. Age, gender, and education were included as covariates in each model to account for their potential influence on the outcome variables. This approach facilitated the estimation of the unique effect of zodiac sign on well-being while controlling for potential confounding variables. For comparison purposes, separate analyses of variance (ANOVAs) were conducted, excluding the covariates. These ANOVAs included each dependent variable and the single independent variable of zodiac sign. This approach provided estimates of the effects of zodiac sign on the dependent variables unadjusted for covariates. Due to the potential for non-normal data distribution in some of the outcome variables, particularly those with limited response categories, the Kruskal–Wallis test was also employed as a complementary nonparametric analysis alongside the ANOVAs and ANCOVAs. This test is used to assess group differences without assuming normality in the data distribution (Coolican, 2017).

In this study, the primary focus for statistical inference was placed on effect sizes rather than p -values due to the substantial sample sizes involved in each analysis. Effect sizes are statistical measures that help evaluate the magnitude and practical significance of relationships between variables, offering insights into the practical significance of each predictor beyond the dichotomous results provided by p -values (Durlak, 2009). To assess the strength of the relationships between zodiac sign and dependent variables, three effect size measures were employed: eta-squared (η^2), partial eta-squared (η^2_p), and omega-squared (ω^2). These are effect size measures commonly used in ANOVA and ANCOVA to assess the magnitude of the relationship between independent and dependent variables. These effect sizes quantify the ratio of variability observed in the scores of the dependent variable that can be attributed to variations in the levels of the independent variable. Eta-squared (η^2) represents the proportion of total variance in the dependent variable accounted for by the independent variable. Partial eta-squared (η^2_p),³ on the other hand, adjusts for the influence of other covariates included in the model (Howitt & Cramer, 2020). Lastly, omega-squared (ω^2) is an alternative measure that corrects for the bias introduced by sample size (Kroes & Finley, 2023).

3 | RESULTS

Table 3 shows the descriptive statistics for all the variables included in the analysis. Figure 1 shows the distribution of these variables across zodiac signs. Table 4 presents the results of ANCOVA models examining the association between zodiac sign and eight outcome variables, controlling for age, gender, and education. For seven of the

³Partial eta-squared (η^2_p) was only used in ANCOVAs, as ANOVAs dealt with a single independent variable, rendering the calculation of η^2_p unnecessary.

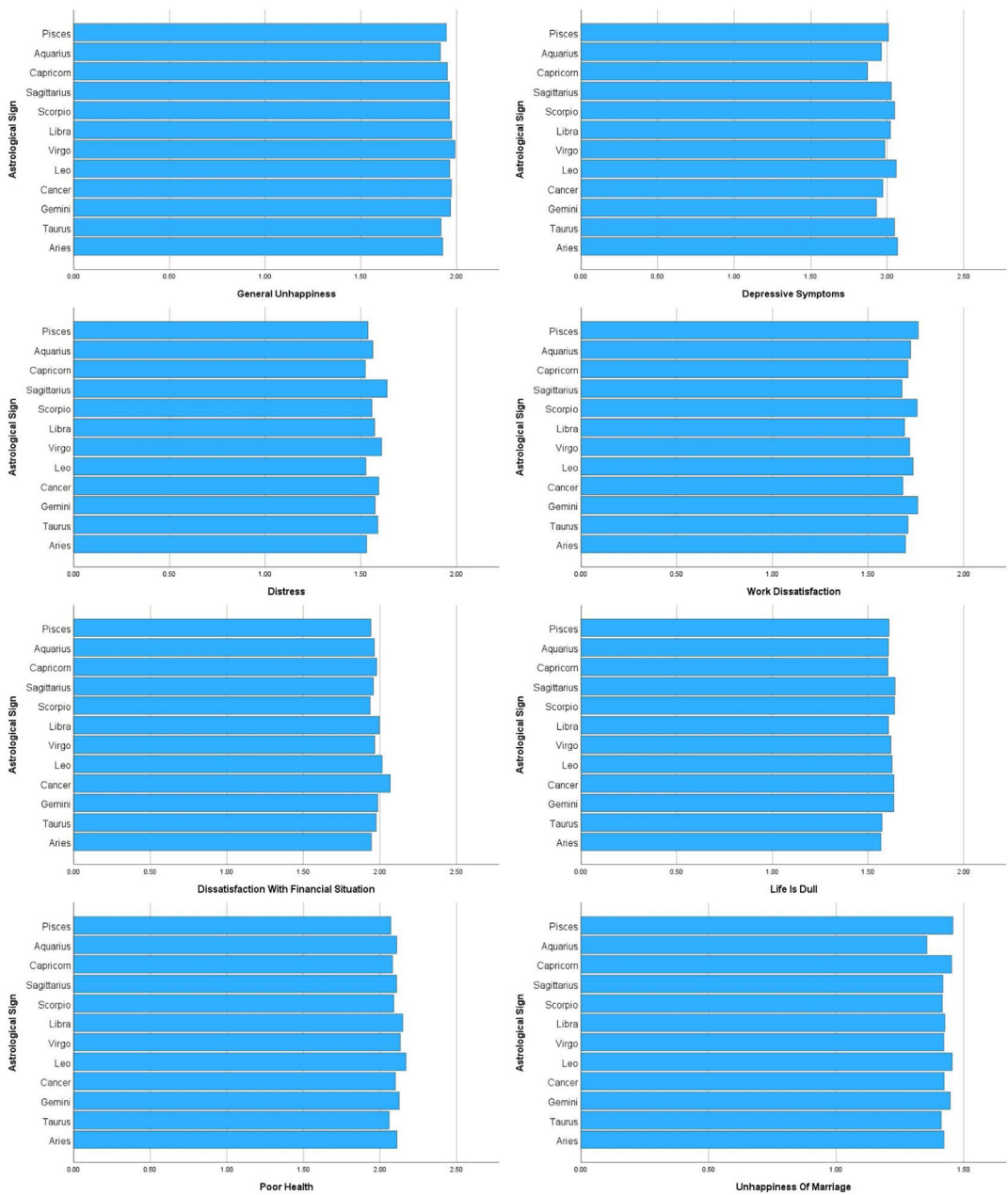


FIGURE 1 Average well-being scores by astrological sign. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

variables, zodiac sign did not have a statistically significant effect on the outcome variables, with all p -values greater than .05. However, zodiac sign was found to be a statistically significant predictor of financial dissatisfaction, which may be due in part to the very large sample size used. The results were consistent with those of the Kruskal–Wallis tests. Nonetheless, the effect sizes were consistently negligible across all outcomes, accounting for 0.3% or less of the variance in the outcomes. Given the very small effect sizes for the eight outcomes, these findings do not provide evidence for an association between an individual's zodiac sign and subjective well-being in this particular population. The analysis was repeated using ANOVAs excluding the covariates. As shown in Table 5, these analyses yielded

TABLE 4 ANCOVA results (with covariates).

Outcome	N	df	F	p	η^2	η^2_p	ω^2	Kruskal–Wallis test		
								Statistic	df	p
General unhappiness	11,845	11 - 11,830	1.155	0.313	0.001	0.001	0.000	13.541	11	0.259
Depressive symptoms	1052	11 - 1037	0.295	0.987	0.003	0.003	0.000	2.389	11	0.997
Distress	1800	11 - 1785	0.192	0.998	0.001	0.001	0.000	3.377	11	0.985
Work dissatisfaction	8106	11 - 8091	1.061	0.389	0.001	0.001	0.000	10.604	11	0.477
Dissatisfaction with financial situation	11,846	11 - 11,831	2.002	0.024	0.002	0.002	0.000	26.721	11	0.005
Life dullness	7866	11 - 7851	0.833	0.607	0.001	0.001	0.000	10.558	11	0.481
Poor health	10,190	11 - 10,175	1.544	0.109	0.002	0.002	0.000	16.444	11	0.125
Unhappiness of marriage	5212	11 - 5197	0.899	0.540	0.002	0.002	0.000	10.369	11	0.497

TABLE 5 ANOVA results (no covariates).

Outcome	N	df	F	p	η^2	ω^2	Kruskal–Wallis test		
							Statistic	df	p
General unhappiness	11,949	11 - 11,937	1.266	0.237	0.001	0.000	14.154	11	0.225
Depressive symptoms	1062	11 - 1050	0.299	0.986	0.003	0.000	2.705	11	0.994
Distress	1822	11 - 1810	0.384	0.963	0.002	0.000	4.053	11	0.968
Work dissatisfaction	8183	11 - 8171	0.923	0.517	0.001	0.000	10.740	11	0.465
Dissatisfaction with financial situation	11,948	11 - 11,936	2.434	0.005	0.002	0.001	26.798	11	0.005
Life dullness	7935	11 - 7923	1.061	0.389	0.001	0.000	11.921	11	0.370
Poor health	10,293	11 - 10,281	1.432	0.151	0.002	0.000	17.241	11	0.101
Unhappiness of marriage	5257	11 - 5245	0.990	0.453	0.002	0.000	11.255	11	0.422

nearly identical results, indicating that the conclusions remain consistent whether or not age, gender, and education are controlled for.

3.1 | Post hoc analysis

To investigate whether astrological sign has predictive validity for well-being outcomes beyond chance and to get a better sense of effect sizes, a random variable was created as a comparative baseline. Random numbers were generated using the R programming language. Specifically, participants were randomly assigned a number between 1 and 12 while ensuring equal sample sizes across the 12 groups. Eight ANOVAs were conducted using the random variable as the sole predictor for well-being measures. The results, presented in Table 6, show striking similarities between the predictive power of astrological sign and the random variable. The random variable did not significantly predict any of the well-being outcomes. Furthermore, the effect sizes associated with the astrological sign and the random variable were negligible and very similar in magnitude. These findings cast considerable doubt on the ability of astrological signs to predict individual differences in well-being. This strongly suggests that knowing someone's astrological sign is just as uninformative about their level of subjective well-being as randomly assigning them a number.

TABLE 6 ANOVA results with a randomly generated predictor.

Outcome	F	p	η^2	ω^2	Kruskal–Wallis test	
					Statistic	p
General unhappiness	1.379	0.175	0.001	0.000	14.945	0.185
Depressive symptoms	0.932	0.508	0.009	0.000	10.486	0.487
Distress	0.940	0.500	0.005	0.000	10.161	0.516
Work dissatisfaction	0.843	0.596	0.001	0.000	8.510	0.667
Dissatisfaction with financial situation	0.317	0.983	0.000	0.000	3.508	0.982
Life dullness	1.193	0.285	0.002	0.000	12.471	0.329
Poor health	1.233	0.259	0.001	0.000	16.330	0.129
Unhappiness of marriage	1.083	0.370	0.002	0.000	12.103	0.356

4 | DISCUSSION

This study aimed to investigate potential relationships between Western astrological zodiac signs and different aspects of subjective well-being using a large, nationally representative sample of American adults. No robust associations were found between individuals' zodiac signs and their self-reported levels of health satisfaction, general happiness, relationship satisfaction, financial satisfaction, job satisfaction, presence of depressive symptoms, or general psychological distress, with or without accounting for respondents' age, gender, and education levels. The effect sizes were nearly zero across all the well-being outcomes examined, indicating that knowing someone's zodiac sign would not provide meaningful information for predicting their well-being levels. These results align with past empirical research which has generally failed to find evidence supporting astrological claims about personality traits or marriage outcomes (e.g., Helgertz & Scott, 2020; Lu et al., 2020; McGrew & McFall, 1990).

The results contradict astrological claims that zodiac signs shape an individual's tendencies, destiny, and susceptibility to feeling happy or successful. Even if only a small percentage of these claims were true, one would expect to see at least some small yet non-trivial effects. Additionally, the absence of any practically significant connection between zodiac signs and well-being casts doubt on the operation of the two previously outlined mechanisms, namely, “self-fulfilling influence” and “astrological stereotyping.” Finally, the post hoc analysis revealed that the predictive power of zodiac signs is statistically indistinguishable from a randomly generated categorical variable. These results indicate that consulting astrological signs tells us just as little about a person's level of well-being as simply putting them into a category based on a coin flip or rolling dice.

Given the lack of a discernible impact of astrological signs on various aspects of well-being, it is crucial to examine the underlying causes of the widespread prevalence of such beliefs among individuals. A variety of psychological factors contribute to the acceptance of astrological beliefs, despite the lack of empirical evidence supporting their validity. For example, humans inherently have pattern-seeking and agency-attributing tendencies (Grayling, 2011; Van Elk, 2013), which leads individuals to perceive associations between astrological signs, personal characteristics, and life outcomes. In addition, the innate desire to understand one's environment and exert control over life events may motivate people to turn to astrology for guidance. This quest for self-understanding and control provides a sense of comfort and reassurance in the midst of life's inherent unpredictability. Relevant psychological models, such as the meaning maintenance model (Heine et al., 2006) and the compensatory control model (Kay et al., 2009), postulate that individuals may adopt belief systems, including pseudoscientific beliefs such as astrology, as a means of maintaining a sense of meaning and perceived control, particularly in times of uncertainty or existential challenge. However, the lack of empirical evidence supporting astrological claims raises questions about the long-term viability and adaptability of such belief systems for promoting well-being.

This study is not without limitations. The use of self-report measures of well-being and a cross-sectional design limit the ability to draw causal inferences or determine the direction of relationships. In addition, the use of single-item measures for most variables and the measurement of a limited number of well-being facets may have affected the reliability and validity of the findings. Future research could address these limitations by using more comprehensive and reliable measures of well-being, employing longitudinal or experimental designs, and considering a broader range of psychological and social aspects of well-being (Joshnanloo, 2021). In addition, the generalizability of the findings may be limited by the use of a single cultural context. Thus, replicating this study across diverse cultural contexts could enhance the generalizability of the findings.

In summary, the present findings undermine astrological claims regarding the impact of zodiac signs on well-being, which has profound implications. These findings underscore the urgent need to increase public awareness of the lack of scientific evidence supporting astrological beliefs through robust public education initiatives. Such initiatives can play a pivotal role in promoting scientific literacy and critical thinking skills, equipping individuals with the rational tools to make informed decisions and challenge pseudoscientific beliefs that lack empirical support. In addition, the pernicious potential of astrological stereotypes to negatively influence perceptions of self and others should also be considered. Multidisciplinary stakeholders (e.g., psychologists, educators, and policy-makers) must work together to educate the public about the risks associated with astrological stereotypes and to promote a more evidence-based understanding of human personality and well-being.

CONFLICT OF INTEREST STATEMENT

The author declares that there is no conflict of interest to report.

DATA AVAILABILITY STATEMENT

The data used in this study are publicly available at <https://gss.norc.oregonstate.edu/get-the-data>.

ORCID

Mohsen Joshnanloo  <https://orcid.org/0000-0001-9350-6219>

REFERENCES

- Clarke, D., Gabriels, T., & Barnes, J. (1996). Astrological signs as determinants of extroversion and emotionality: An empirical study. *The Journal of Psychology*, 130(2), 131–140. <https://doi.org/10.1080/00223980.1996.9914995>
- Coolican, H. (2017). *Research methods and statistics in psychology*. Psychology Press. <https://doi.org/10.4324/9780203769836>
- Das, A., Sharma, M. K., Kashyap, H., & Gupta, S. (2022). Fixating on the future: An overview of increased astrology use. *International Journal of Social Psychiatry*, 68(5), 925–932. <https://doi.org/10.1177/00207640221094155>
- Durlak, J. A. (2009). How to select, calculate, and interpret effect sizes. *Journal of Pediatric Psychology*, 34(9), 917–928. <https://doi.org/10.1093/jpepsy/jsp004>
- Fidrmuc, J., & De Dios Tena, J. (2015). Friday the 13th: The empirics of bad luck. *Kyklos*, 68(3), 317–334. <https://doi.org/10.1111/kykl.12085>
- Grayling, A. C. (2011). Psychology: How we form beliefs. *Nature*, 474(7352), 446–447. <https://doi.org/10.1038/474446a>
- Heine, S. J., Proulx, T., & Vohs, K. D. (2006). The meaning maintenance model: On the coherence of social motivations. *Personality and Social Psychology Review*, 10(2), 88–110. https://doi.org/10.1207/s15327957pspr1002_1
- Helgertz, J., & Scott, K. (2020). The validity of astrological predictions on marriage and divorce: A longitudinal analysis of Swedish register data. *Genus*, 76(1), 34. <https://doi.org/10.1186/s41118-020-00103-5>
- Howitt, D., & Cramer, D. (2020). *Understanding statistics in psychology with SPSS*. Pearson.
- Joshnanloo, M. (2021). Centrality and dimensionality of 14 indicators of mental well-being in four countries: Developing an integrative framework to guide theorizing and measurement. *Social Indicators Research*, 158(2), 727–750. <https://doi.org/10.1007/s11205-021-02723-6>
- Kay, A. C., Whitson, J., Gaucher, D., & Galinsky, A. D. (2009). Compensatory control. *Current Directions in Psychological Science*, 18(5), 264–268. <https://doi.org/10.1111/j.1467-8721.2009.01649.x>
- Kroes, A. D. A., & Finley, J. R. (2023). Demystifying omega squared: Practical guidance for effect size in common analysis of variance designs. *Psychological Methods*. <https://doi.org/10.1037/met0000581>

- Labrot, Z. C., Johnson, C., & Maxime, E. (2022). Horoscopes and emotion. In S. Hupp & R. Wiseman (Eds.), *Investigating pop psychology: Pseudoscience, fringe science, and controversies* (pp. 87–94). Routledge. <https://doi.org/10.4324/9781003107798.10>
- Lu, J. G., Liu, X. L., Liao, H., & Wang, L. (2020). Disentangling stereotypes from social reality: Astrological stereotypes and discrimination in China. *Journal of Personality and Social Psychology*, 119(6), 1359–1379. <https://doi.org/10.1037/pspi0000237>
- McGrew, J. H., & McFall, R. M. (1990). A scientific inquiry into the validity of astrology. *Journal of Scientific Exploration*, 4(1), 75–83.
- Pew Research Center. (2018). 'New age' beliefs common among both religious and nonreligious Americans. <https://www.pewresearch.org/short-reads/2018/10/01/new-age-beliefs-common-among-both-religious-and-nonreligious-americans/>
- Romney, A. C., Harrison, J., & Benson, S. (2023). Looking back to predict the future: A review of empirical support for the self-fulfilling prophecy. *Management Research Review*, 47, 245–260. <https://doi.org/10.1108/mrr-01-2022-0051>
- Tabachnick, B. G., & Fidell, L. S. (2019). *Using multivariate statistics*. Pearson.
- Van Elk, M. (2013). Paranormal believers are more prone to illusory agency detection than skeptics. *Consciousness and Cognition*, 22(3), 1041–1046. <https://doi.org/10.1016/j.concog.2013.07.004>
- Veno, A., & Pamment, P. (1979). Astrological Factors and Personality: A Southern Hemisphere Replication. *The Journal of Psychology*, 101(1), 73–77. <https://doi.org/10.1080/00223980.1979.9915054>

How to cite this article: Joshanloo, M. (2024). The sun's position at birth is unrelated to subjective well-being: Debunking astrological claims. *Kyklos*, 1–10. <https://doi.org/10.1111/kykl.12395>